

Novelty Analysis

Analysis of Novelty, Transience, and Resonance

PENPAL Project

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0.1 Summary

- Humans produce higher novelty (surprisal) and higher transience than the model, indicating stronger deviations from earlier story context.
- Both agents display a positive novelty–resonance relationship (innovation bias), but the slope is steeper for humans.
- This indicates that surprising human turns exert greater forward influence on narrative development.
- AI contributions show lower variance and lower informational impact relative to human inputs.
- Humans are the primary drivers of narrative innovation and thematic redirection.

1 1. Data Loading and Preprocessing

1.1 1.1 Load Data

```
# Load Field Data - using standardized column names
df_res_field <- read_csv(here("data/human-ai/processed/novelty_scores.csv")) %>%
  rename(
    user = author_1,
    ai = author_2,
```

```

    user_surprise = author_1_surprise,
    ai_surprise = author_2_surprise,
    user_transience = author_1_transience,
    ai_transience = author_2_transience
  ) %>%
  filter(!is.na(user_surprise) & !is.na(ai_surprise)) %>%
  mutate(
    user_resonance = user_surprise - user_transience,
    ai_resonance = ai_surprise - ai_transience,
    datatype = "Field"
  ) %>%
  drop_na(user, ai) %>%
  filter(conversation_id != "conv_0bb56093-3033-4615-bb70-ebfa4135589a", conversation_id != "conv_0f18b")

```

1.2 1.2 Reshape Data

```

# Compute word counts for each agent's turn
df_res_field <- df_res_field %>%
  mutate(
    user_nwords = str_count(user, "\\S+"),
    ai_nwords = str_count(ai, "\\S+")
  )

df_res_long <- df_res_field %>%
  pivot_longer(
    cols = c(user_surprise, user_transience, user_nwords,
              ai_surprise, ai_transience, ai_nwords),
    names_to = c("agent", "metric"),
    names_pattern = "(user|ai)_(surprise|transience|nwords)"
  ) %>%
  pivot_wider(names_from = metric, values_from = value) %>%
  mutate(
    resonance = surprise - transience,
    agent = factor(agent, levels = c("user", "ai"), labels = c("User", "LLM"))
  )

```

1.3 1.3 Descriptive Statistics

```

desc_stats <- df_res_long %>%
  group_by(agent) %>%
  summarise(
    n = n(),
    mean_surprise = mean(surprise, na.rm = TRUE),
    sd_surprise = sd(surprise, na.rm = TRUE),
    mean_transience = mean(transience, na.rm = TRUE),
    sd_transience = sd(transience, na.rm = TRUE),
    mean_resonance = mean(resonance, na.rm = TRUE),
    sd_resonance = sd(resonance, na.rm = TRUE)
  )

print(desc_stats)

```

```
## # A tibble: 2 x 8
```

```
## agent      n mean_surprise sd_surprise mean_transience sd_transience
## <fct> <int>      <dbl>      <dbl>          <dbl>          <dbl>
## 1 User      767      -1.82      0.701          -1.02          0.447
## 2 LLM       767      -2.64      0.681          -0.566         0.432
## # i 2 more variables: mean_resonance <dbl>, sd_resonance <dbl>
knitr::kable(desc_stats, caption = "Descriptive Statistics for Novelty Metrics by Agent", digits = 3)
```

Table 1: Descriptive Statistics for Novelty Metrics by Agent

agent	n	mean_surprise	sd_surprise	mean_transience	sd_transience	mean_resonance	sd_resonance
User	767	-1.818	0.701	-1.022	0.447	-0.796	0.875
LLM	767	-2.640	0.681	-0.566	0.432	-2.048	0.807

```
agent_colors <- c("User" = "#377EB8", "LLM" = "#FF7F00")

df_metric_distributions <- df_res_long %>%
  select(agent, surprise, transience, resonance) %>%
  pivot_longer(
    cols = c(surprise, transience, resonance),
    names_to = "metric",
    values_to = "value"
  ) %>%
  mutate(
    metric = factor(
      metric,
      levels = c("surprise", "transience", "resonance"),
      labels = c("Novelty", "Transience", "Resonance")
    )
  )

metric_summary <- df_metric_distributions %>%
  group_by(metric, agent) %>%
  summarise(mean_value = mean(value, na.rm = TRUE), .groups = "drop")

make_metric_distribution <- function(metric_label, show_legend = FALSE) {
  metric_data <- df_metric_distributions %>%
    filter(metric == metric_label)

  metric_lines <- metric_summary %>%
    filter(metric == metric_label)

  density_plot <- ggplot(metric_data, aes(x = value, fill = agent, color = agent)) +
    geom_density(alpha = 0.35, linewidth = 0.9, na.rm = TRUE) +
    geom_vline(
      data = metric_lines,
      aes(xintercept = mean_value, color = agent),
      linetype = "dashed",
      linewidth = 0.8,
      show.legend = FALSE
    ) +
    scale_fill_manual(values = agent_colors, name = "") +
    scale_color_manual(values = agent_colors, name = "") +
```

```

labs(title = metric_label, x = NULL, y = "Density") +
theme(
  legend.position = if (show_legend) "top" else "none",
  plot.title = element_text(hjust = 0.5, size = 16, face = "bold"),
  axis.text.x = element_blank(),
  axis.ticks.x = element_blank(),
  plot.margin = margin(8, 8, 0, 8)
)

box_plot <- ggplot(metric_data, aes(x = value, y = agent, fill = agent)) +
  geom_boxplot(width = 0.45, alpha = 0.7, outlier.alpha = 0.25, na.rm = TRUE) +
  scale_fill_manual(values = agent_colors, name = "") +
  labs(x = metric_label, y = NULL) +
  theme(
    legend.position = "none",
    axis.text.y = element_text(size = 11),
    plot.margin = margin(0, 8, 10, 8)
  )

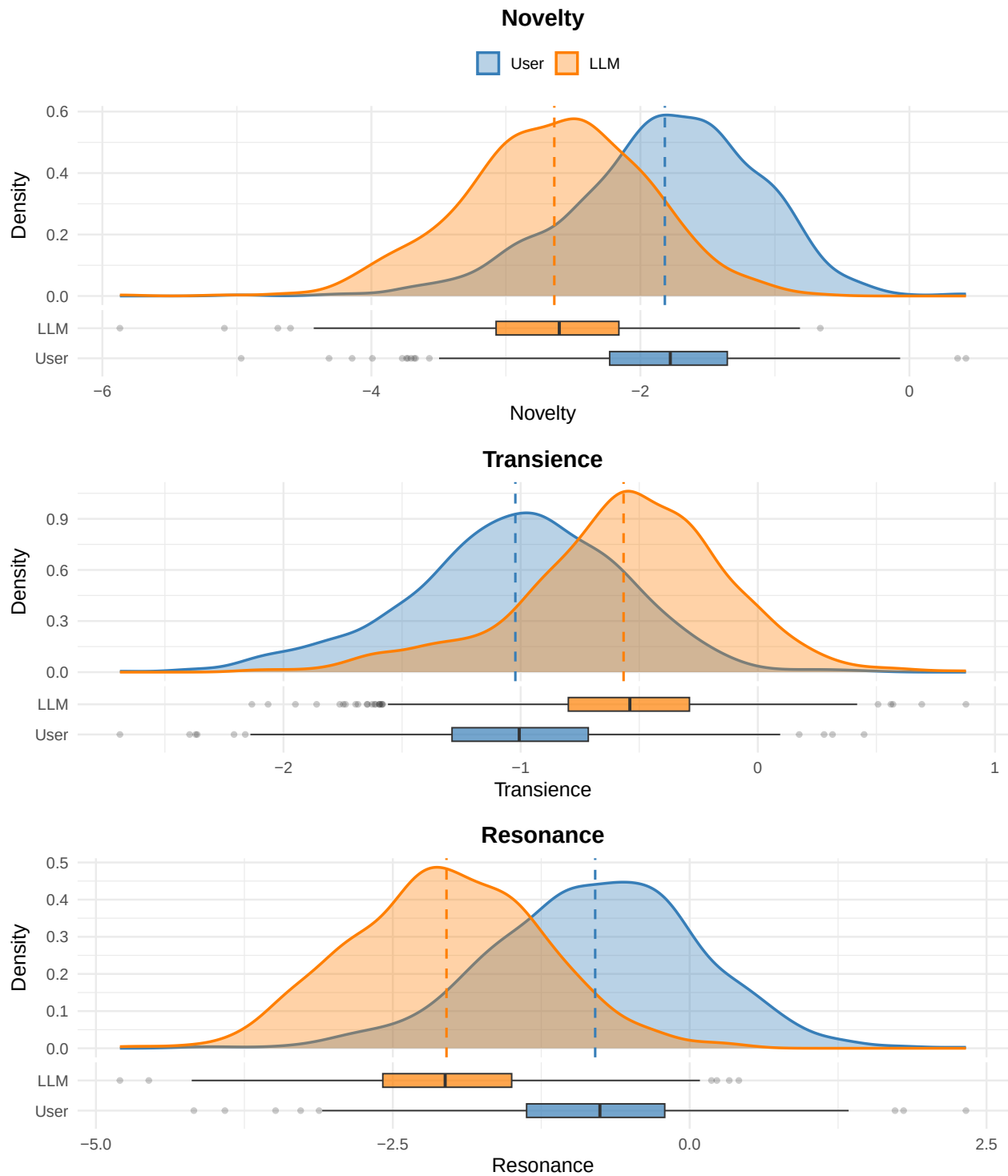
wrap_plots(density_plot, box_plot, ncol = 1, heights = c(3, 1))
}

p_metric_distributions <- wrap_plots(
  make_metric_distribution("Novelty", show_legend = TRUE),
  make_metric_distribution("Transience"),
  make_metric_distribution("Resonance"),
  ncol = 1,
  heights = c(1, 1, 1)
) +
  plot_annotation(
    title = "Distributions of Novelty, Transience, and Resonance by Agent",
    theme = theme(plot.title = element_text(hjust = 0.5, size = 18, face = "bold"))
  )

print(p_metric_distributions)

```

Distributions of Novelty, Transience, and Resonance by Agent



```
ggsave(  
  file.path(GRAPH_DIR, "Novelty_Transience_Resonance_Distributions_Agent.png"),  
  p_metric_distributions,  
  width = 10,  
  height = 12,  
  dpi = 300
```

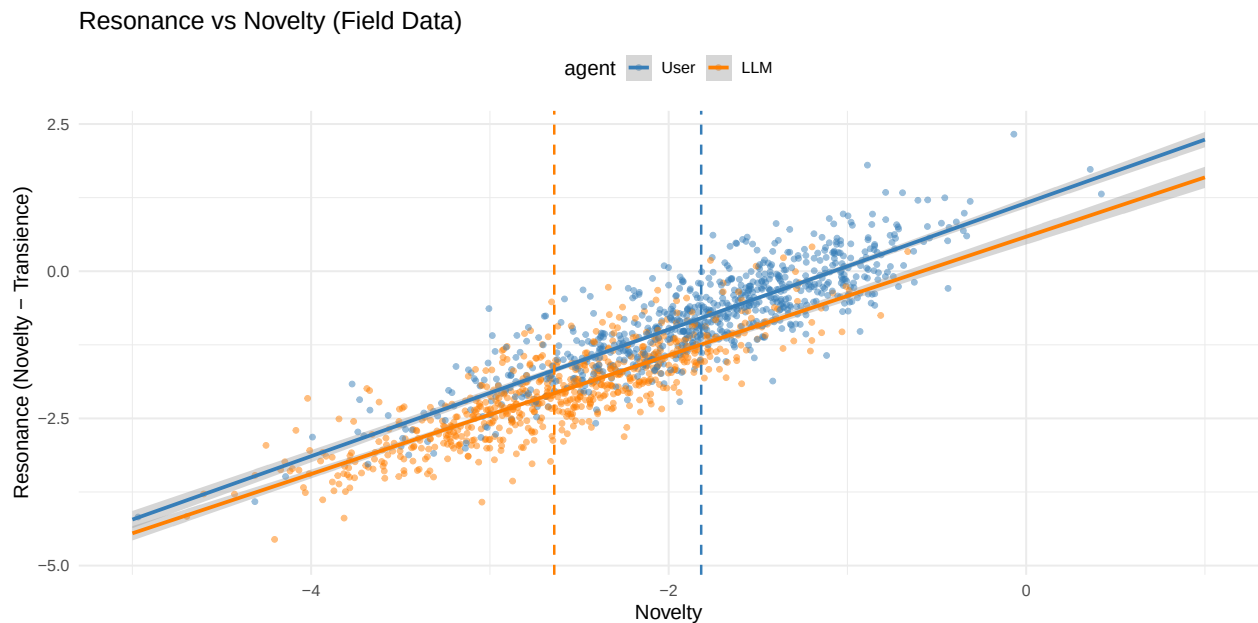
)

2 2. Novelty and Resonance Analysis

2.1 2.1 Resonance vs Novelty Plot

```
# Plot Resonance vs Novelty (Field Data)
p_resonance <- ggplot(df_res_long, aes(x = surprise, y = resonance, color = agent)) +
  geom_point(alpha = 0.5, size = 1.5) +
  geom_smooth(method = "lm", se = TRUE, fullrange = TRUE, size = 1.2) +
  scale_color_manual(values = c("User" = "#377EB8", "LLM" = "#FF7F00")) +
  geom_vline(data = desc_stats,
             aes(xintercept = mean_surprise, color = agent),
             linetype = "dashed", linewidth = 0.8, show.legend = FALSE) +
  labs(
    title = "Resonance vs Novelty (Field Data)",
    x = "Novelty",
    y = "Resonance (Novelty - Transience)"
  ) +
  theme_minimal(base_size = 14) +
  theme(legend.position = "top") + xlim(-5, 1)

print(p_resonance)
```



```
ggsave(file.path(GRAPHS_DIR, "novelty_resonance_vs_novelty.png"),
        p_resonance, width = 12, height = 6, dpi = 300)
```

2.2 2.2 Transience vs Novelty Plot

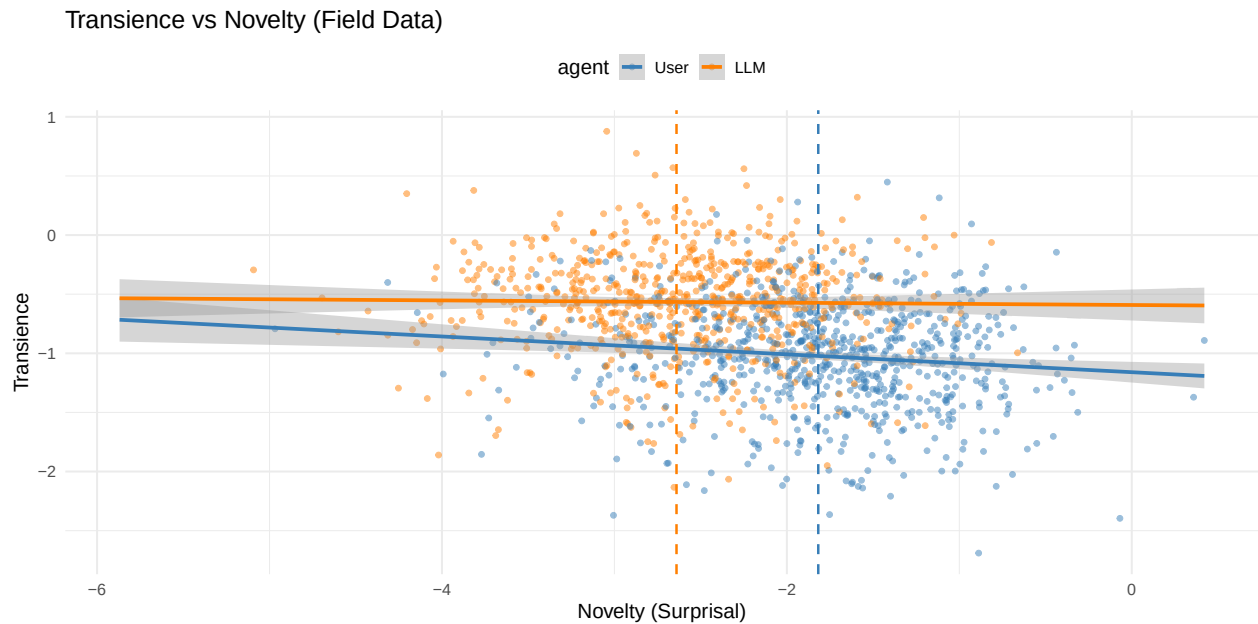
```
p_transience <- ggplot(df_res_long, aes(x = surprise, y = transience, color = agent)) +
  geom_point(alpha = 0.5, size = 1.5) +
  geom_smooth(method = "lm", se = TRUE, fullrange = TRUE, size = 1.2) +
```

```

scale_color_manual(values = c("User" = "#377EB8", "LLM" = "#FF7F00")) +
  geom_vline(data = desc_stats,
    aes(xintercept = mean_surprise, color = agent),
    linetype = "dashed", linewidth = 0.8, show.legend = FALSE) +
  labs(
    title = "Transience vs Novelty (Field Data)",
    x = "Novelty (Surprisal)",
    y = "Transience"
  ) +
  theme_minimal(base_size = 14) +
  theme(legend.position = "top")

print(p_transience)

```



```

ggsave(file.path(GRAPHS_DIR, "novelty_transience_vs_novelty.png"),
  p_transience, width = 12, height = 6, dpi = 300)

```

3. Statistical Models

3.1 Resonance Model

```

# Model without length control
model_resonance <- lmer(resonance ~ surprise * agent + (1 | conversation_id), data = df_res_long)
cat("=== Resonance Model ===\n")

```

```
## === Resonance Model ===
```

```
print(summary(model_resonance)$coefficients)
```

```
##               Estimate Std. Error      df    t value      Pr(>|t|)
## (Intercept)    1.21843067 0.04554494 1135.317  26.752275 1.180453e-122
## surprise       1.10789626 0.02242379 1429.730  49.407191 1.627895e-311
## agentLLM      -0.51501098 0.07999238 1427.903  -6.438250 1.646645e-10
## surprise:agentLLM -0.05528472 0.03345474 1428.208  -1.652523 9.864780e-02
```

```

cat("\n=== Slope Comparison (Resonance) ===\n")

##
## === Slope Comparison (Resonance) ===
print(emtrends(model_resonance, specs = "agent", var = "surprise"))

## agent surprise.trend      SE    df lower.CL upper.CL
## User          1.11 0.0224 1430      1.06      1.15
## LLM           1.05 0.0252 1434      1.00      1.10
##
## Degrees-of-freedom method: kenward-roger
## Confidence level used: 0.95

# Model WITH length control
model_resonance_len <- lmer(resonance ~ surprise * agent + nwords + (1 | conversation_id), data = df_res)
cat("\n=== Resonance Model (controlling for turn length) ===\n")

##
## === Resonance Model (controlling for turn length) ===
print(summary(model_resonance_len)$coefficients)

##              Estimate Std. Error      df    t value      Pr(>|t|)
## (Intercept)   1.129046654 0.080002370  990.5451 14.112665 2.410673e-41
## surprise      1.104138525 0.022744560 1423.1627 48.545170 3.672747e-304
## agentLLM     -0.536142108 0.081728249 1419.5108 -6.560059 7.515289e-11
## nwords        0.003219824 0.002282028  836.2298  1.410948 1.586318e-01
## surprise:agentLLM -0.058077118 0.033510812 1420.8084 -1.733086 8.329736e-02

cat("\n=== Slope Comparison (Resonance, length-controlled) ===\n")

##
## === Slope Comparison (Resonance, length-controlled) ===
print(emtrends(model_resonance_len, specs = "agent", var = "surprise"))

## agent surprise.trend      SE    df lower.CL upper.CL
## User          1.10 0.0228 1425      1.059      1.15
## LLM           1.05 0.0260 1428      0.995      1.10
##
## Degrees-of-freedom method: kenward-roger
## Confidence level used: 0.95

```

3.2 3.2 Transience Model

```

# Model without length control
model_transience <- lmer(transience ~ surprise * agent + (1 | conversation_id), data = df_res_long)

cat("=== Transience Model ===\n")

## === Transience Model ===
print(summary(model_transience)$coefficients)

##              Estimate Std. Error      df    t value      Pr(>|t|)
## (Intercept)  -1.21843066 0.04554494 1135.317 -26.752275 1.180444e-122
## surprise     -0.10789626 0.02242379 1429.730  -4.811688 1.655055e-06

```

```

## agentLLM          0.51501099 0.07999238 1427.903    6.438250 1.646644e-10
## surprise:agentLLM 0.05528472 0.03345474 1428.208    1.652523 9.864780e-02

cat("\n=== Slope Comparison (Transience) ===\n")

##
## === Slope Comparison (Transience) ===

print(emtrends(model_transience, specs = "agent", var = "surprise"))

## agent surprise.trend      SE    df lower.CL upper.CL
## User          -0.1079 0.0224 1430   -0.152 -0.06386
## LLM           -0.0526 0.0252 1434   -0.102 -0.00316
##
## Degrees-of-freedom method: kenward-roger
## Confidence level used: 0.95

# Model WITH length control
model_transience_len <- lmer(transience ~ surprise * agent + nwords + (1 | conversation_id), data = df_)

cat("\n=== Transience Model (controlling for turn length) ===\n")

##
## === Transience Model (controlling for turn length) ===

print(summary(model_transience_len)$coefficients)

##              Estimate Std. Error      df    t value      Pr(>|t|)
## (Intercept)   -1.129046653 0.080002370  990.5451 -14.112665 2.410674e-41
## surprise      -0.104138525 0.022744560 1423.1627  -4.578612 5.089017e-06
## agentLLM       0.536142108 0.081728249 1419.5108   6.560059 7.515289e-11
## nwords        -0.003219824 0.002282028  836.2298  -1.410948 1.586318e-01
## surprise:agentLLM 0.058077118 0.033510812 1420.8084   1.733086 8.329736e-02

cat("\n=== Slope Comparison (Transience, length-controlled) ===\n")

##
## === Slope Comparison (Transience, length-controlled) ===

print(emtrends(model_transience_len, specs = "agent", var = "surprise"))

## agent surprise.trend      SE    df lower.CL upper.CL
## User          -0.1041 0.0228 1425   -0.149 -0.0595
## LLM           -0.0461 0.0260 1428   -0.097  0.0049
##
## Degrees-of-freedom method: kenward-roger
## Confidence level used: 0.95

```

4 4. Supplementary: Resonance vs Novelty by LLM Type

```

desc_stats_llm <- df_res_long %>%
  filter(!is.na(llm_type)) %>%
  group_by(agent, llm_type) %>%
  summarise(mean_surprise = mean(surprise, na.rm = TRUE), .groups = "drop")

p_llm_user <- df_res_long %>%
  filter(!is.na(llm_type), agent == "User") %>%

```

```

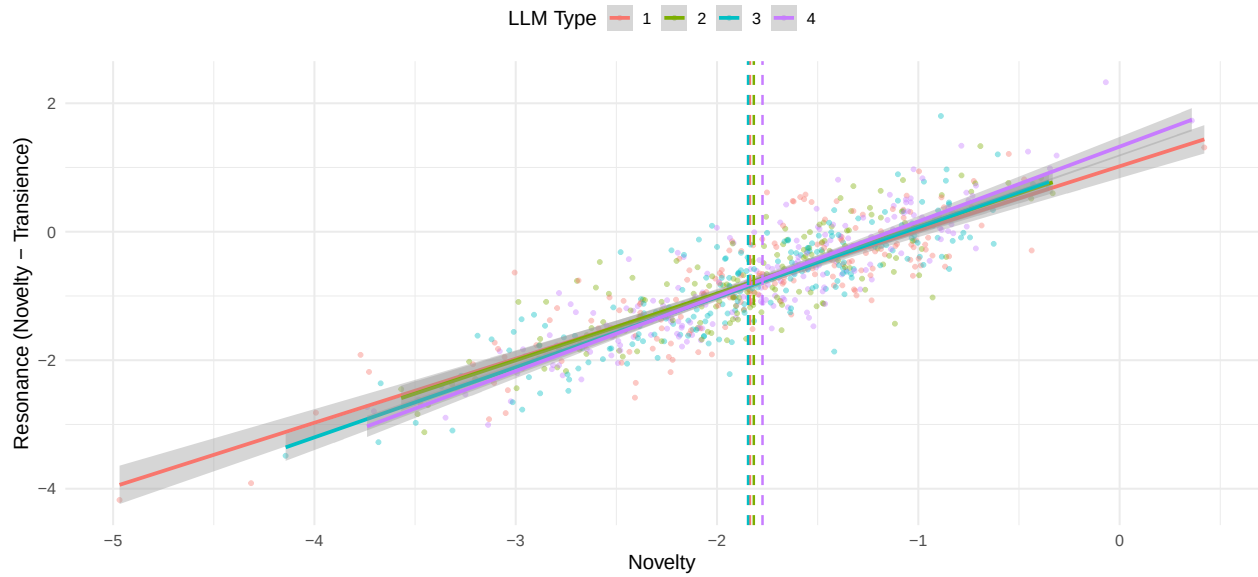
ggplot(aes(x = surprise, y = resonance, color = factor(llm_type))) +
geom_point(alpha = 0.4, size = 1.2) +
geom_smooth(method = "lm", se = TRUE, linewidth = 1.2) +
geom_vline(data = desc_stats_llm %>% filter(agent == "User"),
           aes(xintercept = mean_surprise, color = factor(llm_type)),
           linetype = "dashed", linewidth = 0.8, show.legend = FALSE) +
labs(
  title = "User: Resonance vs Novelty by LLM Type",
  x = "Novelty",
  y = "Resonance (Novelty - Transience)",
  color = "LLM Type"
) +
theme_minimal(base_size = 14) +
theme(legend.position = "top",
      plot.title = element_text(hjust = 0.5, face = "bold"))

p_llm_ai <- df_res_long %>%
  filter(!is.na(llm_type), agent == "LLM") %>%
  ggplot(aes(x = surprise, y = resonance, color = factor(llm_type))) +
  geom_point(alpha = 0.4, size = 1.2) +
  geom_smooth(method = "lm", se = TRUE, linewidth = 1.2) +
  geom_vline(data = desc_stats_llm %>% filter(agent == "LLM"),
            aes(xintercept = mean_surprise, color = factor(llm_type)),
            linetype = "dashed", linewidth = 0.8, show.legend = FALSE) +
  labs(
    title = "LLM: Resonance vs Novelty by LLM Type",
    x = "Novelty",
    y = "Resonance (Novelty - Transience)",
    color = "LLM Type"
  ) +
  theme_minimal(base_size = 14) +
  theme(legend.position = "top",
        plot.title = element_text(hjust = 0.5, face = "bold"))

print(p_llm_user)

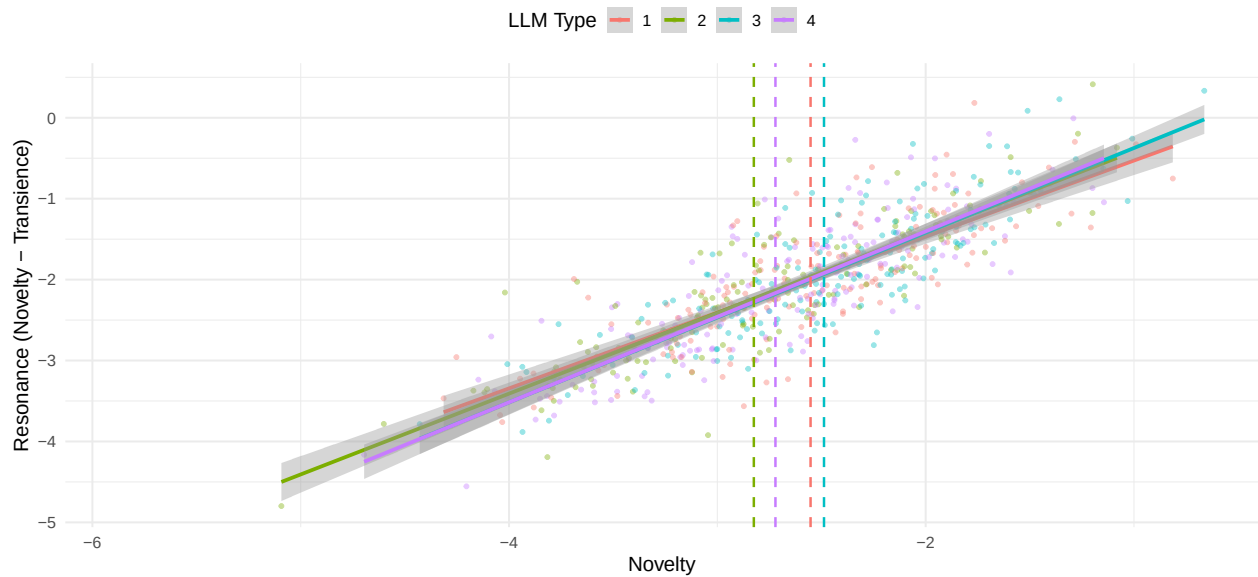
```

User: Resonance vs Novelty by LLM Type



```
print(p_llm_ai)
```

LLM: Resonance vs Novelty by LLM Type



```
ggsave(file.path(GRAPHS_DIR, "novelty_resonance_llm_type_user.png"),
        p_llm_user, width = 12, height = 6, dpi = 300)
ggsave(file.path(GRAPHS_DIR, "novelty_resonance_llm_type_ai.png"),
        p_llm_ai, width = 12, height = 6, dpi = 300)
```

5 5. Session Information

```
sessionInfo()
```

```
## R version 4.4.2 (2024-10-31)
```

```

## Platform: aarch64-apple-darwin20
## Running under: macOS 26.3
##
## Matrix products: default
## BLAS:   /Library/Frameworks/R.framework/Versions/4.4-arm64/Resources/lib/libRblas.0.dylib
## LAPACK: /Library/Frameworks/R.framework/Versions/4.4-arm64/Resources/lib/libRlapack.dylib; LAPACKv
##
## locale:
## [1] en_US.UTF-8/en_US.UTF-8/en_US.UTF-8/C/en_US.UTF-8/en_US.UTF-8
##
## time zone: Australia/Melbourne
## tzcode source: internal
##
## attached base packages:
## [1] stats      graphics  grDevices  utils      datasets  methods    base
##
## other attached packages:
## [1] emmeans_1.10.6 here_1.0.1      afex_1.4-1      cowplot_1.1.3
## [5] patchwork_1.3.0 zoo_1.8-12      stringi_1.8.4   arrow_19.0.1.1
## [9] lmerTest_3.1-3 lme4_1.1-35.5  Matrix_1.7-1    lubridate_1.9.3
## [13] forcats_1.0.0 stringr_1.5.1   dplyr_1.1.4     purrr_1.0.2
## [17] readr_2.1.5    tidyr_1.3.1     tibble_3.2.1    ggplot2_3.5.1
## [21] tidyverse_2.0.0
##
## loaded via a namespace (and not attached):
## [1] tidyselect_1.2.1 farver_2.1.2      fastmap_1.2.0
## [4] TH.data_1.1-2     pacman_0.5.1      digest_0.6.37
## [7] timechange_0.3.0  estimability_1.5.1 lifecycle_1.0.4
## [10] survival_3.7-0    magrittr_2.0.3    compiler_4.4.2
## [13] rlang_1.1.4       tools_4.4.2       utf8_1.2.4
## [16] yaml_2.3.10       knitr_1.50        labeling_0.4.3
## [19] bit_4.0.5         plyr_1.8.9        abind_1.4-8
## [22] multcomp_1.4-26   withr_3.0.2       numDeriv_2016.8-1.1
## [25] grid_4.4.2        fansi_1.0.6       xtable_1.8-4
## [28] colorspace_2.1-1 scales_1.3.0      MASS_7.3-61
## [31] tinytex_0.52      cli_3.6.3         mvtnorm_1.3-3
## [34] rmarkdown_2.28    crayon_1.5.3      ragg_1.3.2
## [37] generics_0.1.3    rstudioapi_0.16.0 reshape2_1.4.4
## [40] tzdb_0.4.0        minqa_1.2.8       splines_4.4.2
## [43] assertthat_0.2.1  parallel_4.4.2    vctrs_0.6.5
## [46] boot_1.3-31       sandwich_3.1-1    carData_3.0-5
## [49] car_3.1-3         hms_1.1.3         pbkrtest_0.5.3
## [52] bit64_4.0.5       Formula_1.2-5     systemfonts_1.1.0
## [55] glue_1.7.0        nloptr_2.1.1      codetools_0.2-20
## [58] gtable_0.3.5      munsell_0.5.1     pillar_1.9.0
## [61] htmltools_0.5.8.1 R6_2.5.1          textshaping_0.4.0
## [64] rprojroot_2.0.4   vroom_1.6.5       evaluate_1.0.3
## [67] lattice_0.22-6    backports_1.5.0   broom_1.0.6
## [70] Rcpp_1.0.13       coda_0.19-4.1     nlme_3.1-166
## [73] mgcv_1.9-1        xfun_0.56         pkgconfig_2.0.3

```

Analysis complete. All figures saved to `analysis/figures/` and tables to `analysis/tables/`.